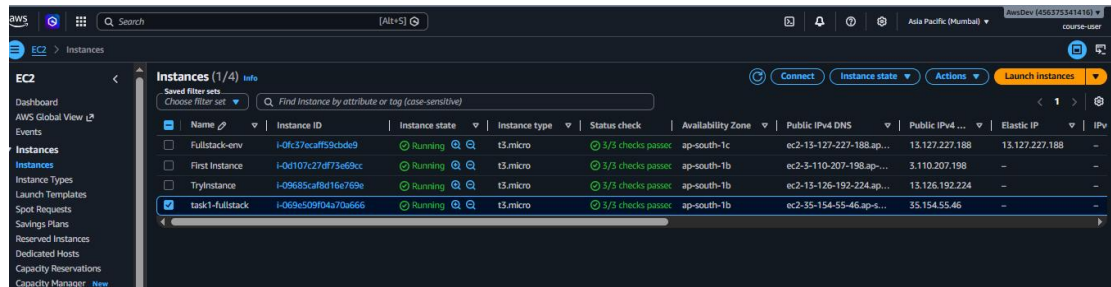


# Module-9

## Assignment-6

### 1. Deploy Your flask backend and express frontend in amazon single ec2 instance.

Single EC2 Instance for the deployment of Frontend and Backend.



Instance connected with the terminal using SSH URL :

```
ubuntu@ip-172-31-12-194: ~/FullStack-Assignment
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop
$ ssh -i "fulltask1.pem" ubuntu@ec2-35-154-55-46.ap-south-1.compute.amazonaws.com
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue Aug 11 17:11:33 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   89.9% of 6.61GB Processes:      130
Memory usage: 49%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.12.194

=> / is using 89.9% of 6.61GB

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

*** System restart required ***
Last login: Tue Aug 11 17:05:15 2026 from 1.187.172.131
ubuntu@ip-172-31-12-194:~$ cd FullStack_Assignment
-bash: cd: FullStack_Assignment: No such file or directory
ubuntu@ip-172-31-12-194:~$ cd FullStack-Assignment
ubuntu@ip-172-31-12-194:~/FullStack-Assignment$ ls
backend  docker-compose.yml  frontend
ubuntu@ip-172-31-12-194:~/FullStack-Assignment$ cat docker-compose.yml
version: "3.8"

services:
  backend:
    build: ./backend
    container_name: flask_backend
    ports:
      - "5000:5000"

  frontend:
    build: ./Frontend
    container_name: express_frontend
    ports:
      - "3000:3000"
    depends_on:
      - backendubuntu@ip-172-31-12-194:~/FullStack-Assignment$ |
```

Inbound Rules create for the Backend with the Port No.5000 and and Frontend with the Port No.3000

DetailsStatus and alarmsMonitoringSecurityNetworkingStorageTags

▼ Security details

IAM role

-

Security groups

sg-024ede08947a01002 (launch-wizard-4)

Owner ID

456375341416

Launch time

Tue Aug 11 2026 19:43:18 GMT+0530 (India Standard Time)

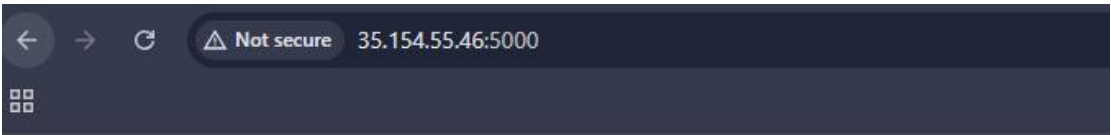
▼ Inbound rules

Q Filter rules

| Name | Security group rule ID | Port range | Protocol | Source    | Security groups                 | Description |
|------|------------------------|------------|----------|-----------|---------------------------------|-------------|
| -    | sgr-0cf7e58574eb5c917  | 5000       | TCP      | 0.0.0.0/0 | <a href="#">launch-wizard-4</a> | -           |
| -    | sgr-0744018b1b663a1dd  | 3000       | TCP      | 0.0.0.0/0 | <a href="#">launch-wizard-4</a> | -           |
| -    | sgr-05d58de750a8410d4  | 22         | TCP      | 0.0.0.0/0 | <a href="#">launch-wizard-4</a> | -           |

▼ Outbound rules

Backend Working



BackEnd is Running

Frontend Workng



User Registration Form

Name:

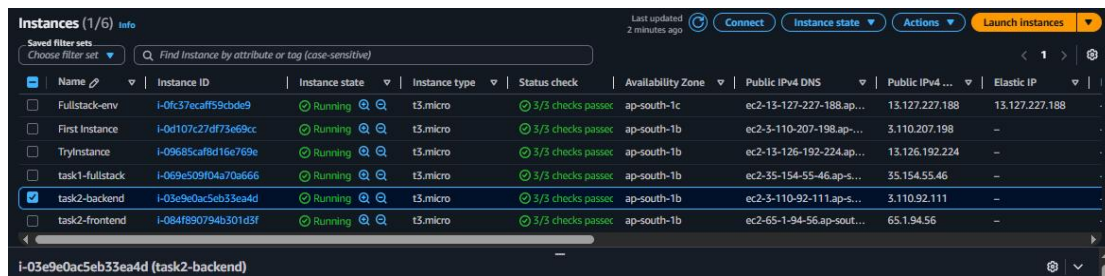
Email:

Age:

Submit

## 2. Deploy Your flask backend and express frontend in separate ec2 instances.

Separate EC2 created for both Frontend and Backend



The screenshot shows the AWS Management Console 'Instances' page. It lists several EC2 instances, including 'task2-backend' which is highlighted. The table below represents the data visible in the screenshot.

| Name            | Instance ID         | Instance state | Instance type | Status check      | Availability Zone | Public IPv4 DNS           | Public IPv4 ... | Elastic IP     |
|-----------------|---------------------|----------------|---------------|-------------------|-------------------|---------------------------|-----------------|----------------|
| Fullstack-env   | i-0fc37ecaff59cbde9 | Running        | t3.micro      | 3/3 checks passed | ap-south-1c       | ec2-13-127-227-188.ap...  | 13.127.227.188  | 13.127.227.188 |
| First Instance  | i-0d107c27df73e69cc | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-3-110-207-198.ap...   | 3.110.207.198   | -              |
| TryInstance     | i-09685caf8d16e769e | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-13-126-192-224.ap...  | 13.126.192.224  | -              |
| task1-fullstack | i-069e509f04a70a666 | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-35-154-55-46.ap-s...  | 35.154.55.46    | -              |
| task2-backend   | i-03e9e0ac5eb33ea4d | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-3-110-92-111.ap-s...  | 3.110.92.111    | -              |
| task2-frontend  | i-084f890794b301d3f | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-65-1-94-56.ap-sout... | 65.1.94.56      | -              |

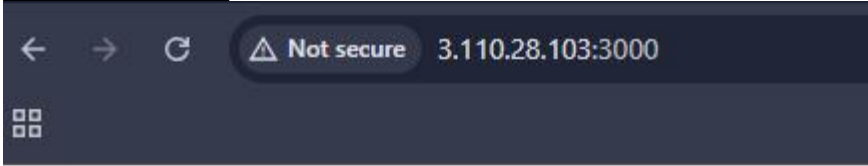
Frontend EC2 connected with the terminal

```
ubuntu@ip-172-31-8-185: ~  
LENOVO@DESKTOP-9084F2Q MINGW64 ~/Desktop  
$ ssh -i "Front.pem" ubuntu@ec2-65-1-94-56.ap-south-1.compute.amazonaws.com  
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)  
  
* Documentation:  https://docs.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/pro  
  
System information as of Tue Aug 11 17:38:53 UTC 2026  
  
System load:  0.0           Temperature:   -273.1 C  
Usage of /:   30.7% of 6.61GB Processes:     114  
Memory usage: 26%          Users logged in: 0  
Swap usage:   0%           IPv4 address for ens5: 172.31.8.185  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
  
Last login: Tue Aug 11 17:30:41 2026 from 1.187.172.131  
ubuntu@ip-172-31-8-185:~$ |
```

Docker image created

```
Successfully tagged express-frontend:latest
ubuntu@ip-172-31-8-185:~/FullStack-Assignment/frontend$ sudo docker images
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
express-frontend:latest             de5a68507f36                      1.59GB      401MB
node:20                             8f693eaa7e0a                      1.59GB      415MB
ubuntu@ip-172-31-8-185:~/FullStack-Assignment/frontend$ sudo docker run -d --name
e express_frontend -p 3000:3000 express-frontend
ac1ba84a67086e7946bfa68e30e90aa25510ad1c7db2aff74caf0517f3b620e8
ubuntu@ip-172-31-8-185:~/FullStack-Assignment/frontend$ sudo docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS
PORTS
ac1ba84a6708   express-frontend                    "docker-entrypoint.s..." 13 seconds ago Up 1
2 seconds     0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
express_frontend
ubuntu@ip-172-31-8-185:~/FullStack-Assignment/frontend$ |
```

Frontend Working



User Registration Form

Name:

Email:

Age:

Submit

EC2 for Backend

| Instances (1/6) Info                               |                 |                     |                |               |                   |                   |                           |                 |                |
|----------------------------------------------------|-----------------|---------------------|----------------|---------------|-------------------|-------------------|---------------------------|-----------------|----------------|
| Last updated 2 minutes ago                         |                 |                     |                |               |                   |                   |                           |                 |                |
| Save filter sets                                   |                 |                     |                |               |                   |                   |                           |                 |                |
| Choose filter set                                  |                 |                     |                |               |                   |                   |                           |                 |                |
| Find Instance by attribute or tag (case-sensitive) |                 |                     |                |               |                   |                   |                           |                 |                |
|                                                    | Name            | Instance ID         | Instance state | Instance type | Status check      | Availability Zone | Public IPv4 DNS           | Public IPv4 ... | Elastic IP     |
| <input type="checkbox"/>                           | Fullstack-env   | i-0fc37ecaff59cbde9 | Running        | t3.micro      | 3/3 checks passed | ap-south-1c       | ec2-13-127-227-188.ap...  | 13.127.227.188  | 13.127.227.188 |
| <input type="checkbox"/>                           | First Instance  | i-0d107c27df73e69cc | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-3-110-207-198.ap...   | 3.110.207.198   | -              |
| <input type="checkbox"/>                           | TryInstance     | i-09685caf8d16e769e | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-13-126-192-224.ap...  | 13.126.192.224  | -              |
| <input type="checkbox"/>                           | task1-fullstack | i-069e509f04a70a666 | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-35-154-55-46.ap-s...  | 35.154.55.46    | -              |
| <input checked="" type="checkbox"/>                | task2-backend   | i-03e9e0ac5eb33ea4d | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-3-110-92-111.ap-s...  | 3.110.92.111    | -              |
| <input type="checkbox"/>                           | task2-frontend  | i-084f890794b301d3f | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-65-1-94-56.ap-sout... | 65.1.94.56      | -              |



## Backend build Images

```
ubuntu@ip-172-31-12-115: ~/FullStack-Assignment/backend
Step 5/6 : EXPOSE 5000
--> Running in d1966fd7b1a0
--> Removed intermediate container d1966fd7b1a0
--> 2ee14ac59417
Step 6/6 : CMD ["python", "app.py"]
--> Running in 9837a17a7f07
--> Removed intermediate container 9837a17a7f07
--> 6c7f2d3125cd
Successfully built 6c7f2d3125cd
Successfully tagged flask-backend:latest
ubuntu@ip-172-31-12-115:~/FullStack-Assignment/backend$ sudo docker images

```

| IMAGE                | ID           | DISK USAGE | CONTENT SIZE | EXTRA |
|----------------------|--------------|------------|--------------|-------|
| flask-backend:latest | 6c7f2d3125cd | 1.62GB     | 416MB        |       |
| python:3.11          | d0199e2a90bf | 1.61GB     | 427MB        |       |

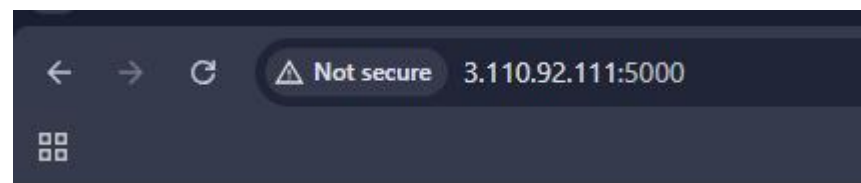
```
ubuntu@ip-172-31-12-115:~/FullStack-Assignment/backend$ sudo docker run -d --name flask_backend -p 5000:5000 flask-backend
cbe4bb8ce243baa812eb3d7a6d08993c4e34b6ca9eff34524890da4788120650
ubuntu@ip-172-31-12-115:~/FullStack-Assignment/backend$ sudo docker ps

```

| CONTAINER ID | IMAGE         | COMMAND         | CREATED        | STATUS        |
|--------------|---------------|-----------------|----------------|---------------|
| cbe4bb8ce243 | flask-backend | "python app.py" | 13 seconds ago | Up 13 seconds |

```
0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp flask_backend
ubuntu@ip-172-31-12-115:~/FullStack-Assignment/backend$ |
```

## Backend Working

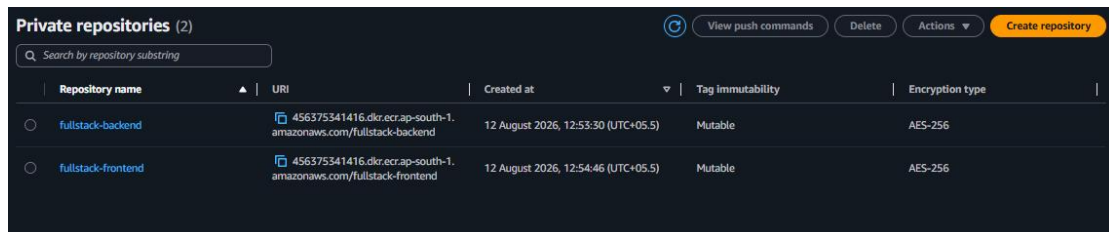


BackEnd is Running

### 3. Deploy Your flask backend and express frontend Docker Container using aws ecr, ecs and vpc services

--> Express Frontend

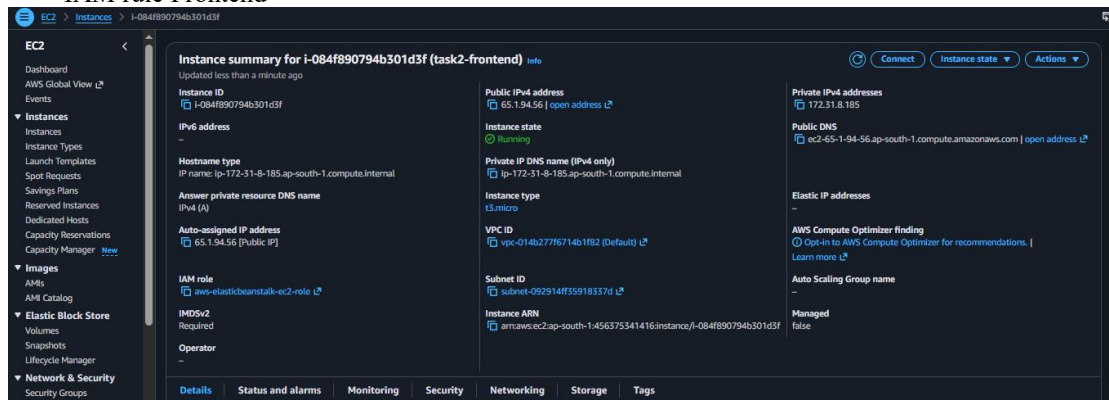
- repo created



Private repositories (2)

| Repository name    | URI                                                              | Created at                          | Tag immutability | Encryption type |
|--------------------|------------------------------------------------------------------|-------------------------------------|------------------|-----------------|
| fullstack-backend  | 456375341416.dkr.ecr.ap-south-1.amazonaws.com/fullstack-backend  | 12 August 2026, 12:53:30 (UTC+05.5) | Mutable          | AES-256         |
| fullstack-frontend | 456375341416.dkr.ecr.ap-south-1.amazonaws.com/fullstack-frontend | 12 August 2026, 12:54:46 (UTC+05.5) | Mutable          | AES-256         |

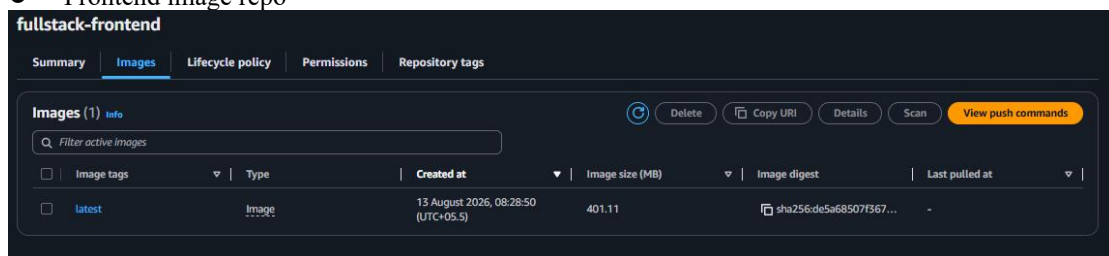
- IAM rule Frontend



Instance summary for i-084f890794b301d3f (task2-frontend)

|                                                                              |                                                                                         |                                                                                                           |
|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Instance ID</b><br>i-084f890794b301d3f                                    | <b>Public IPv4 address</b><br>65.1.94.56   open address                                 | <b>Private IPv4 addresses</b><br>172.31.8.185                                                             |
| <b>IPv6 address</b><br>-                                                     | <b>Instance state</b><br>Running                                                        | <b>Public DNS</b><br>ec2-65-1-94-56.ap-south-1.compute.amazonaws.com   open address                       |
| <b>Hostname type</b><br>IP name: ip-172-31-8-185.ap-south-1.compute.internal | <b>Private IP DNS name (IPv4 only)</b><br>ip-172-31-8-185.ap-south-1.compute.internal   | <b>Elastic IP addresses</b><br>-                                                                          |
| <b>Answer private resource DNS name</b><br>IPv4 (A)                          | <b>Instance type</b><br>t3.micro                                                        | <b>AWS Compute Optimizer finding</b><br>Opt-in to AWS Compute Optimizer for recommendations.   Learn more |
| <b>Auto-assigned IP address</b><br>65.1.94.56 (public IP)                    | <b>VPC ID</b><br>vpc-014b277f6714b1f82 (Default)                                        | <b>Auto Scaling Group name</b><br>-                                                                       |
| <b>IAM role</b><br>aws-elasticbeanstalk-ec2-role                             | <b>Subnet ID</b><br>subnet-092914f35918337d                                             | <b>Managed</b><br>false                                                                                   |
| <b>IMDSv2</b><br>Required                                                    | <b>Instance ARN</b><br>arn:aws:ec2:ap-south-1:456375341416:instance/i-084f890794b301d3f |                                                                                                           |
| <b>Operator</b><br>-                                                         |                                                                                         |                                                                                                           |

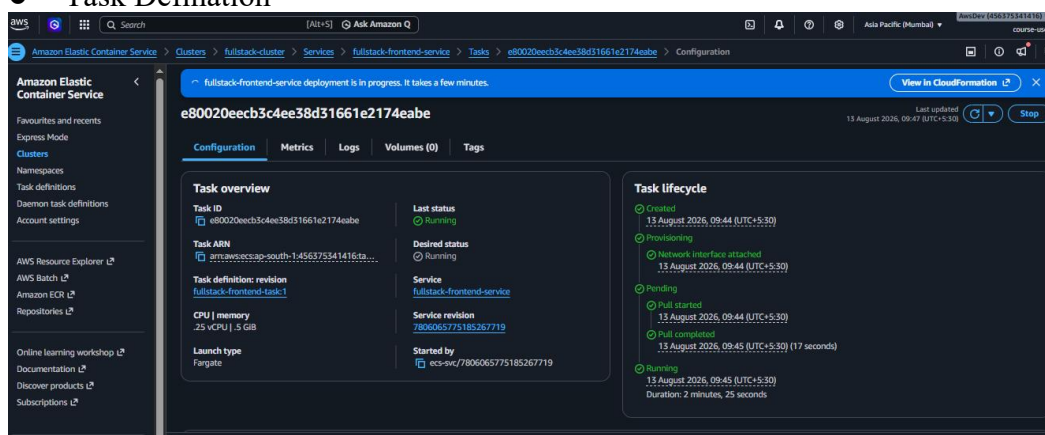
- Frontend image repo



fullstack-frontend

| Image tags | Type  | Created at                          | Image size (MB) | Image digest             | Last pulled at |
|------------|-------|-------------------------------------|-----------------|--------------------------|----------------|
| latest     | Image | 13 August 2026, 08:28:50 (UTC+05.5) | 401.11          | sha256:de5af68507f367... | -              |

- Task Defination

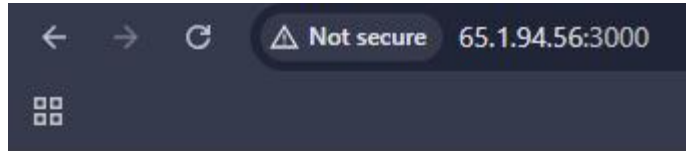


Amazon Elastic Container Service

Task Definition: e80020eebc3c4ee38d31661e2174eabe

| Task overview                                                                                    | Task lifecycle                                                                        |
|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Task ID</b><br>e80020eebc3c4ee38d31661e2174eabe                                               | <b>Created</b><br>13 August 2026, 09:44 (UTC+5:30)                                    |
| <b>Task ARN</b><br>arn:aws:ecs:ap-south-1:456375341416:task-definition/fullstack-frontend-task-1 | <b>Provisioning</b><br>Network interface attached<br>13 August 2026, 09:44 (UTC+5:30) |
| <b>Task definition: revision</b><br>fullstack-frontend-task-1                                    | <b>Pending</b><br>Pull started<br>13 August 2026, 09:44 (UTC+5:30)                    |
| <b>CPU   memory</b><br>25 vCPU   5 GB                                                            | <b>Pull completed</b><br>13 August 2026, 09:45 (UTC+5:30) (17 seconds)                |
| <b>Launch type</b><br>Fargate                                                                    | <b>Running</b><br>13 August 2026, 09:45 (UTC+5:30)<br>Duration: 2 minutes, 25 seconds |

- Frontend Success



## User Registration Form

Name:

Email:

Age:

--> Flask Backend

- repo created

A screenshot of the AWS Private repositories console. It shows a table with two repositories. The first repository is 'fullstack-backend' with URI '456375341416.dkr.ecr.ap-south-1.amazonaws.com/fullstack-backend', created at '12 August 2026, 12:53:30 (UTC+05.5)', tag immutability 'Mutable', and encryption type 'AES-256'. The second repository is 'fullstack-frontend' with URI '456375341416.dkr.ecr.ap-south-1.amazonaws.com/fullstack-frontend', created at '12 August 2026, 12:54:46 (UTC+05.5)', tag immutability 'Mutable', and encryption type 'AES-256'.

| Repository name    | URI                                                              | Created at                          | Tag immutability | Encryption type |
|--------------------|------------------------------------------------------------------|-------------------------------------|------------------|-----------------|
| fullstack-backend  | 456375341416.dkr.ecr.ap-south-1.amazonaws.com/fullstack-backend  | 12 August 2026, 12:53:30 (UTC+05.5) | Mutable          | AES-256         |
| fullstack-frontend | 456375341416.dkr.ecr.ap-south-1.amazonaws.com/fullstack-frontend | 12 August 2026, 12:54:46 (UTC+05.5) | Mutable          | AES-256         |

- IAM Rule Backend

A screenshot of the AWS IAM console. It shows a table of instances with columns: Name, Instance ID, Instance state, Instance type, Status check, Availability Zone, Public IPv4 DNS, Public IPv4..., Elastic IP, and IAM role. The instance 'task2-backend' is highlighted. Below the table, the details for the instance 'I-03e9e0ac5eb33ea4d' are shown, including its IAM role 'EC2-Role', AMI ID 'ami-01a00762f46d584a1', and platform details 'Linux/UNIX'.

| Name            | Instance ID          | Instance state | Instance type | Status check      | Availability Zone | Public IPv4 DNS           | Public IPv4... | Elastic IP     | IAM role |
|-----------------|----------------------|----------------|---------------|-------------------|-------------------|---------------------------|----------------|----------------|----------|
| Fullstack-env   | I-01c37ecaff59bde9   | Running        | t3.micro      | 3/3 checks passed | ap-south-1c       | ec2-13-127-227-188.ap...  | 13.127.227.188 | 13.127.227.188 |          |
| First Instance  | I-0d107c27d773e69cc  | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-3-110-207-198.ap...   | 3.110.207.198  | -              |          |
| TryInstance     | I-09685cafb8d16e769e | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-13-126-192-224.ap...  | 13.126.192.224 | -              |          |
| task1-fullstack | I-069e509f04a70a666  | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-35-154-55-46.ap-s...  | 35.154.55.46   | -              |          |
| task2-backend   | I-03e9e0ac5eb33ea4d  | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-3-110-92-111.ap-s...  | 3.110.92.111   | -              |          |
| task2-frontend  | I-064f890794b301d3f  | Running        | t3.micro      | 3/3 checks passed | ap-south-1b       | ec2-65-1-94-56.ap-sout... | 65.1.94.56     | -              |          |

**I-03e9e0ac5eb33ea4d (task2-backend)**

**IAM role**  
EC2-Role

**AMI ID**  
ami-01a00762f46d584a1

**Subnet ID**  
subnet-092914ff35918337d

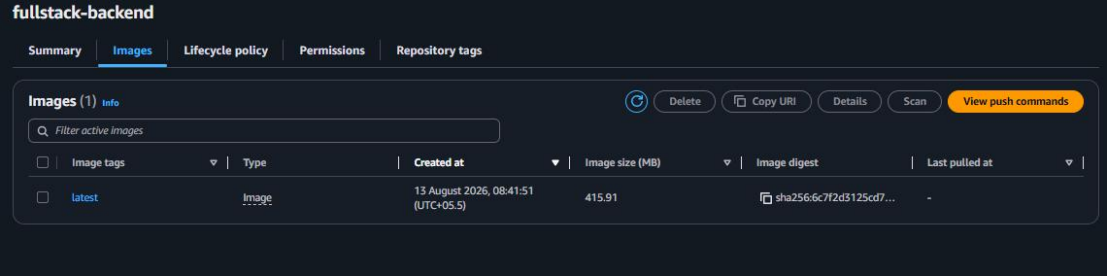
**Instance ARN**  
arn:aws:ec2:ap-south-1:456375341416:instance/i-03e9e0ac5eb33ea4d

**Auto Scaling Group name**  
-

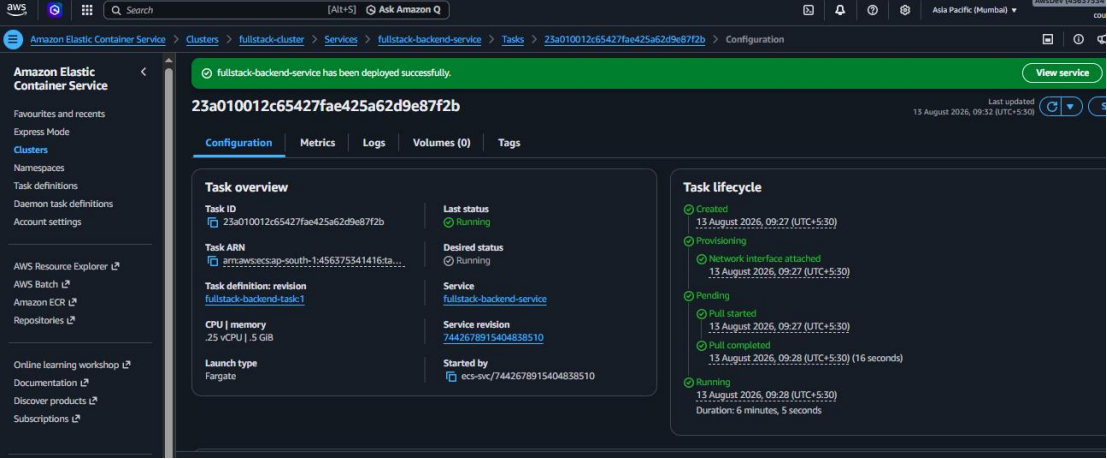
**Managed**  
false

**Platform details**  
Linux/UNIX

● Backend Image Repo



● Task Definition Backend



● Backend Success



BackEnd is Running